

Titration report

LabX

Method

Method ID	CDC_GranTED-DYN	Release state	Released
Name	Titr TRIS with HCl - Gran Verification	Modified by	PredefinedUser
Modified on	2/5/2026 7:31:59 AM		

Summary

Samples

	Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.007000 mL	OK
		Concentration R2	0.10014 mol/L	OK
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.011809 mL	OK
		Concentration R2	0.10024 mol/L	OK

Statistics

R1

Mean	5.009404 mL	Minimum	5.007000 mL
Standard deviation	0.003400 mL	Rel. standard deviation	0.068 %
Total samples	2	Samples excluded	0

R2

Mean	0.10019 mol/L	Minimum	0.10014 mol/L
Standard deviation	0.00007 mol/L	Rel. standard deviation	0.071 %
Total samples	2	Samples excluded	0

Main Results

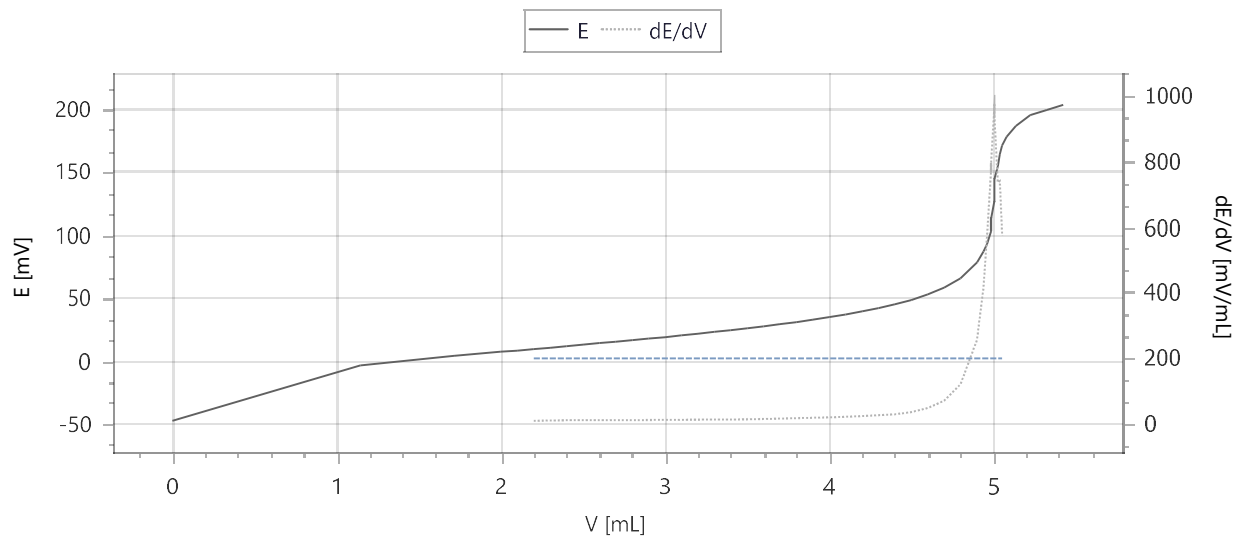
Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.007000 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10014 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.011809 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10024 mol/L	OK

Titration (EQP) [1] Scope 1/1, Sample 1/2

E - V & dE/dV - V Curve Scope 1/1, Sample 1/2

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Measured Values Scope 1/1, Sample 1/2

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-47.4		25.0			
10	1.1430	-3.4	1.1430	25.0	44.0		
16	1.7145	4.3	0.5715	25.0	7.7		
22	2.0000	7.5	0.2855	25.0	3.2		
56	2.1000	8.2	0.1000	25.0	0.7		
60	2.2000	9.6	0.1000	25.0	1.4	10.17	
64	2.3000	10.7	0.1000	25.0	1.1	11.19	
68	2.4000	11.9	0.1000	25.0	1.2	11.82	
72	2.5000	13.1	0.1000	25.0	1.2	12.19	
75	2.6000	14.3	0.1000	25.0	1.2	11.81	
79	2.7000	15.4	0.1000	25.0	1.1	12.00	
83	2.8000	16.8	0.1000	25.0	1.4	12.08	
87	2.9000	17.9	0.1000	25.0	1.1	12.58	
90	3.0000	19.1	0.1000	25.0	1.2	13.17	
93	3.1000	20.5	0.1000	25.0	1.4	13.39	
96	3.2000	21.9	0.1000	25.0	1.4	13.63	
99	3.3000	23.4	0.1000	25.0	1.5	14.16	
103	3.4000	24.7	0.1000	25.0	1.3	14.20	
106	3.5000	26.0	0.1000	25.0	1.3	14.66	
110	3.6000	27.7	0.1000	25.0	1.7	15.68	
113	3.7000	29.3	0.1000	25.0	1.6	16.89	
116	3.8000	30.9	0.1000	25.0	1.6	18.29	
120	3.9000	33.0	0.1000	25.0	2.1	19.48	
124	4.0000	35.0	0.1000	25.0	2.0	20.75	
127	4.1000	37.1	0.1000	25.0	2.1	22.43	
132	4.2000	39.5	0.1000	25.0	2.4	24.53	
136	4.3000	42.1	0.1000	25.0	2.6	27.17	
140	4.4000	45.1	0.1000	25.0	3.0	30.01	
144	4.5000	48.6	0.1000	25.0	3.5	36.50	
149	4.6000	53.0	0.1000	25.0	4.4	48.97	
155	4.7000	58.4	0.1000	25.0	5.4	71.20	

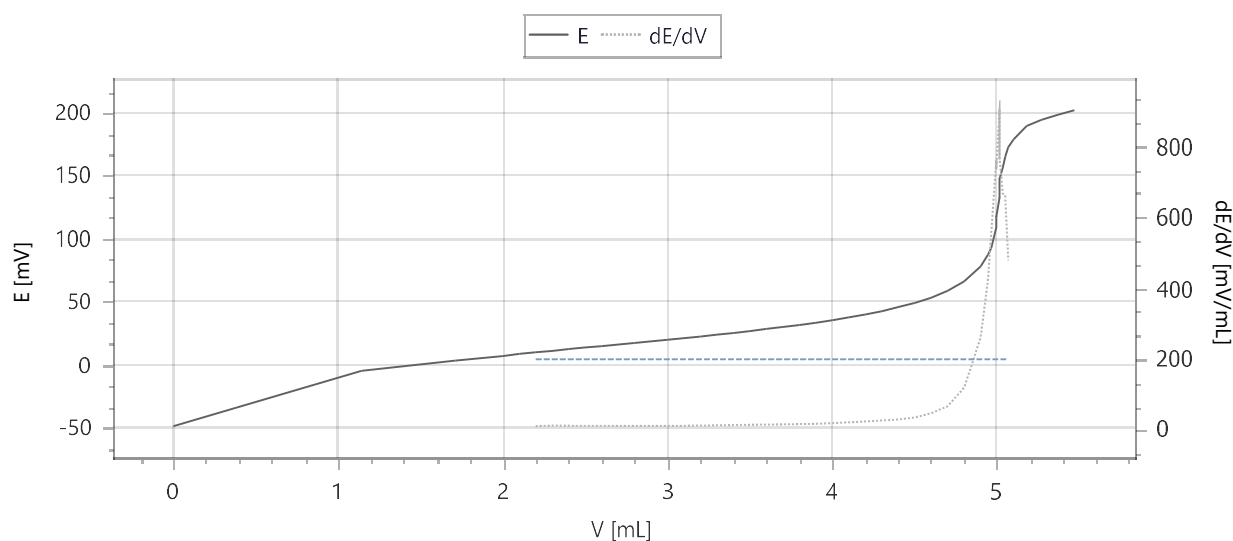
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161	4.8000	65.9	0.1000	25.0	7.5	121.68	
168	4.9000	78.6	0.1000	25.0	12.7	256.58	
178	4.9395	87.1	0.0395	25.0	8.5	415.47	
181	4.9620	93.1	0.0225	25.0	6.0	579.69	
190	4.9855	103.5	0.0235	25.0	10.4	770.54	
194	4.9965	113.7	0.0110	25.0	10.2	792.31	
200	5.0065	127.3	0.0100	25.0	13.6	970.88	
	5.007000	128.2				972.15	EQP
210	5.0165	144.6	0.0100	25.0	17.3	932.17	
213	5.0265	155.0	0.0100	25.0	10.4	735.36	
217	5.0390	164.5	0.0125	25.0	9.5	739.85	
222	5.0535	171.4	0.0145	25.0	6.9	572.87	
228	5.0800	178.2	0.0265	25.0	6.8		
235	5.1370	187.0	0.0570	25.0	8.8		
238	5.2250	195.5	0.0880	25.0	8.5		
246	5.3250	199.7	0.1000	25.0	4.2		
250	5.4250	203.6	0.1000	25.0	3.9		

Titration (EQP) [1] Scope 1/1, Sample 2/2

E - V & dE/dV - V Curve Scope 1/1, Sample 2/2



Measured Values Scope 1/1, Sample 2/2

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-48.6		25.0			
10	1.1430	-4.6	1.1430	25.0	44.0		
16	1.7145	3.4	0.5715	25.0	8.0		
22	2.0000	6.9	0.2855	25.0	3.5		
56	2.1000	8.7	0.1000	25.0	1.8		
60	2.2000	10.1	0.1000	25.0	1.4	11.71	
63	2.3000	11.1	0.1000	25.0	1.0	13.03	
66	2.4000	12.6	0.1000	25.0	1.5	12.87	
70	2.5000	13.8	0.1000	25.0	1.2	12.09	
74	2.6000	14.9	0.1000	25.0	1.1	12.28	

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77	2.7000	16.2	0.1000	25.0	1.3	12.48	EQP
80	2.8000	17.5	0.1000	25.0	1.3	12.21	
84	2.9000	18.7	0.1000	25.0	1.2	12.48	
87	3.0000	19.9	0.1000	25.0	1.2	12.66	
90	3.1000	21.3	0.1000	25.0	1.4	12.65	
94	3.2000	22.4	0.1000	25.0	1.1	13.22	
98	3.3000	23.9	0.1000	25.0	1.5	14.07	
102	3.4000	25.3	0.1000	25.0	1.4	14.78	
106	3.5000	26.7	0.1000	25.0	1.4	15.44	
110	3.6000	28.5	0.1000	25.0	1.8	16.05	
113	3.7000	30.1	0.1000	25.0	1.6	16.55	
116	3.8000	31.7	0.1000	25.0	1.6	17.46	
120	3.9000	33.5	0.1000	25.0	1.8	18.31	
125	4.0000	35.4	0.1000	25.0	1.9	19.95	
129	4.1000	37.6	0.1000	25.0	2.2	22.41	
133	4.2000	39.9	0.1000	25.0	2.3	24.81	
137	4.3000	42.5	0.1000	25.0	2.6	27.63	
140	4.4000	45.8	0.1000	25.0	3.3	30.40	
145	4.5000	49.2	0.1000	25.0	3.4	36.00	
150	4.6000	53.3	0.1000	25.0	4.1	48.10	
155	4.7000	58.8	0.1000	25.0	5.5	67.68	
162	4.8000	66.1	0.1000	25.0	7.3	120.10	
169	4.9000	78.1	0.1000	25.0	12.0	260.70	
177	4.9440	87.6	0.0440	25.0	9.5	421.46	
183	4.9650	92.6	0.0210	25.0	5.0	557.61	
189	4.9950	109.0	0.0300	25.0	16.4	759.95	
199	5.0050	117.5	0.0100	25.0	8.5	738.70	
	5.011809	127.9				908.32	
209	5.0150	132.8	0.0100	25.0	15.3	902.72	
217	5.0250	147.4	0.0100	25.0	14.6	767.60	
220	5.0350	156.5	0.0100	25.0	9.1	667.47	
224	5.0490	165.2	0.0140	25.0	8.7	663.55	
227	5.0680	172.6	0.0190	25.0	7.4	479.18	
235	5.1005	179.1	0.0325	25.0	6.5		
238	5.1800	189.7	0.0795	25.0	10.6		
247	5.2700	194.4	0.0900	25.0	4.7		
253	5.3700	198.6	0.1000	25.0	4.2		
256	5.4700	202.1	0.1000	25.0	3.5		

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1
Sample ID 2

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Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
05.02.2026 08:30:45	PredefinedUser	Administrator